



DEVELOPMENT AND IMPLEMENTATION OF A TRAFFIC ANOMALY DETECTION XAPP (O-RAN) ON 5G-SA TESTBED@ITB

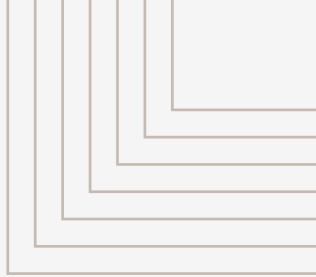
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Background & Research Problem

Background & Motivation

5G Requirements

5G development requires high network capacity, low latency, and massive connectivity. However, conventional Radio Access Network (RAN) infrastructure, which is closed and monolithic, has become a bottleneck due to vendor lock-in. This has driven the industry to shift toward a more open, intelligent, and interoperable architecture: O-RAN.

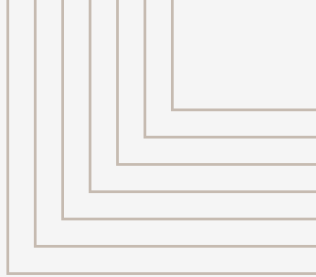
O-RAN Architecture

The O-RAN paradigm, which prioritizes an open and disaggregated architecture, introduces new security risks by expanding the attack surface on open interfaces such as E2.

Attack Vulnerabilities

This openness exposes the network to fatal attacks targeting Availability (service availability), such as Signaling Storms and Denial of Service (DoS), which are difficult to detect because they exploit standard RAN protocols.





Background & Research Problem

Background & Motivation

Why use Machine Learning?

Conventional security methods based on static rules (thresholds) have proven insufficient in handling 5G traffic dynamics because they evaluate individual metrics in isolation rather than between multiple correlated metrics.

Conclusion

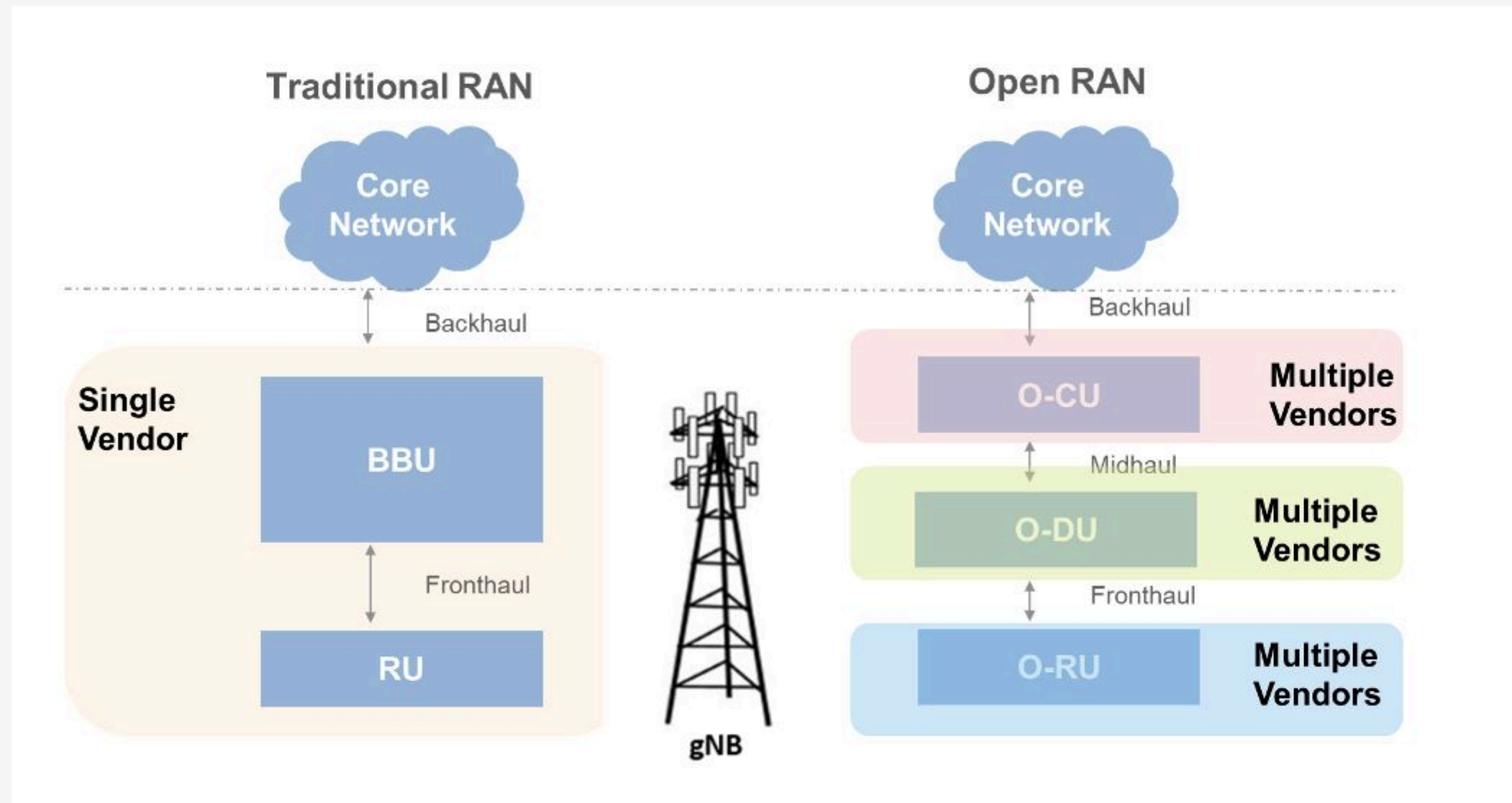
We intend to implement a Machine Learning-based anomaly detection that is capable of learning network behavior patterns and is directly integrated into the Near-RT RIC architecture within a representative testbed environment.

Research Problems:

- 01 How to implement a traffic anomaly detection xApp that can operate effectively within the O-RAN architecture?
- 02 How to integrate this xApp into the Near-RT RIC on the existing O-RAN testbed?

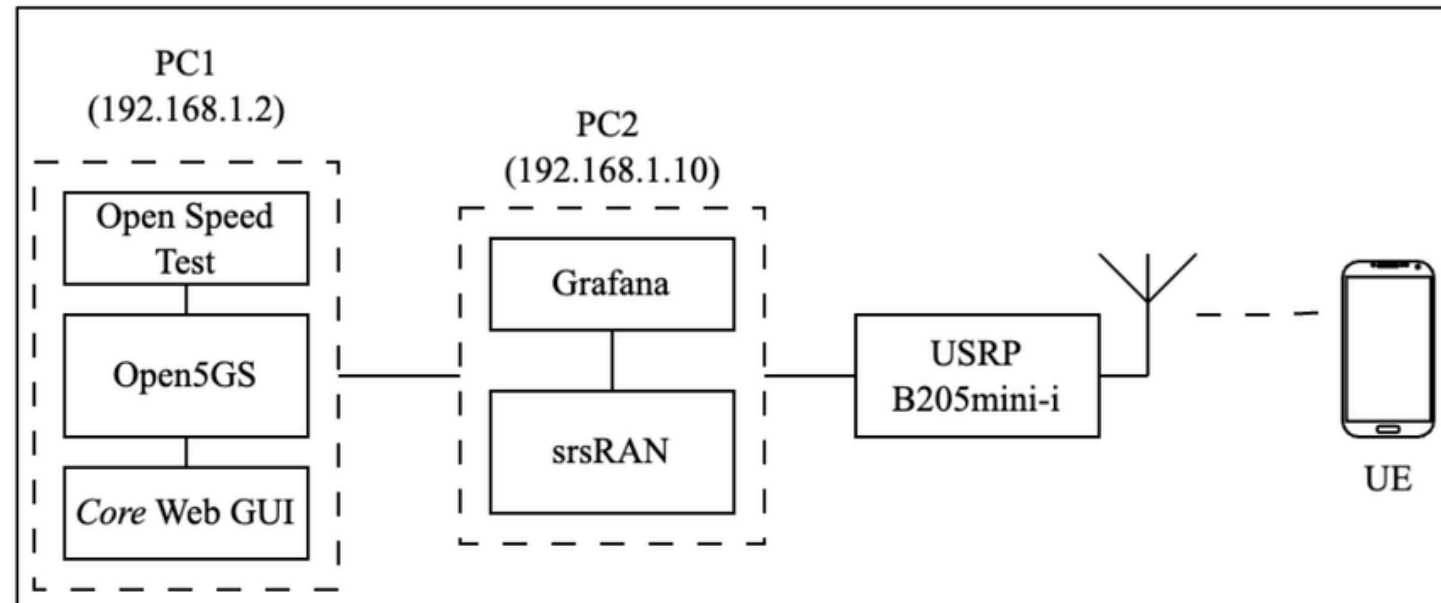


O-RAN Architecture

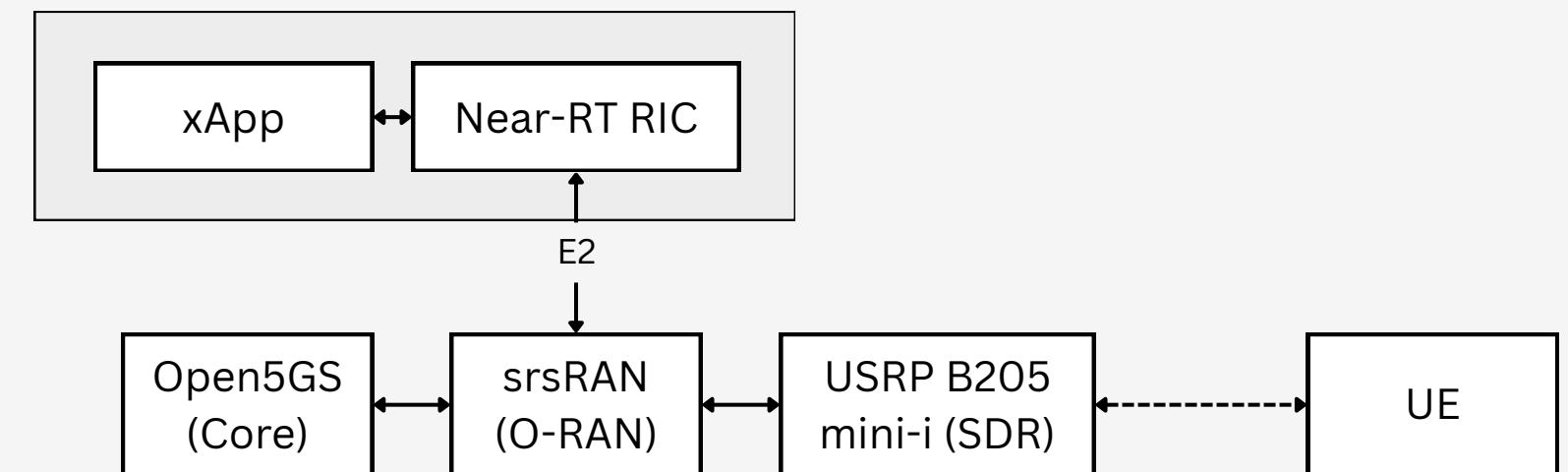


- **Traditional RAN:** Base Band Unit (BBU) and Radio Unit are typically proprietary hardware from a single vendor
- **O-RAN:** BBU is broken into Open Central Unit (O-CU) and Open Distributed Unit (O-DU), they are connected with each other and an Open Radio Unit (O-RU) via standardized open interfaces

Current Architecture



What we want to achieve



Explanation of Terms:

- **Near-RT RIC:** The intelligent controller in the O-RAN architecture that manages radio resources in near-real-time (10ms to 1s) and hosts the xApps.
- **xApp:** A microservice application running on the RIC that executes specific logic
- **RAN:** The radio access part of the network (implemented via srsRAN) that connects user devices to the core network using radio signals.
- **Core:** The network's central brain (Open5GS) responsible for managing SIM authentication, data sessions, and routing traffic to the internet.
- **UE:** The end-user device (Oppo Reno 8) that connects wirelessly to the 5G network to access data services.
- **E2:** The standard open interface that connects the Near-RT RIC to the RAN node, allowing the xApp to collect data and control the network.
- **USRP:** The SDR hardware that functions as the 5G base station to physically transmit and receive radio signals.

Design Objectives & Constraints



Objective 1: System Integration

- Subobjective 1: Integrated with Near-RT RIC and 5G-SA Testbed components
 - Constraint 1: Does not cause crashes, overload, or communication disruptions between the RIC and gNB.
 - Constraint 2: Compatible with srsRAN gNB, Open5GS core, and USRP B205mini-i devices.

Objective 2: Network Anomaly Detection

- Subobjective 2: High Detection Accuracy

Objective 3: Maintain System Performance

- Subobjective 3: High Detection Efficiency
- Subobjective 4: Low Latency



Subobjective Fulfillment Metrics

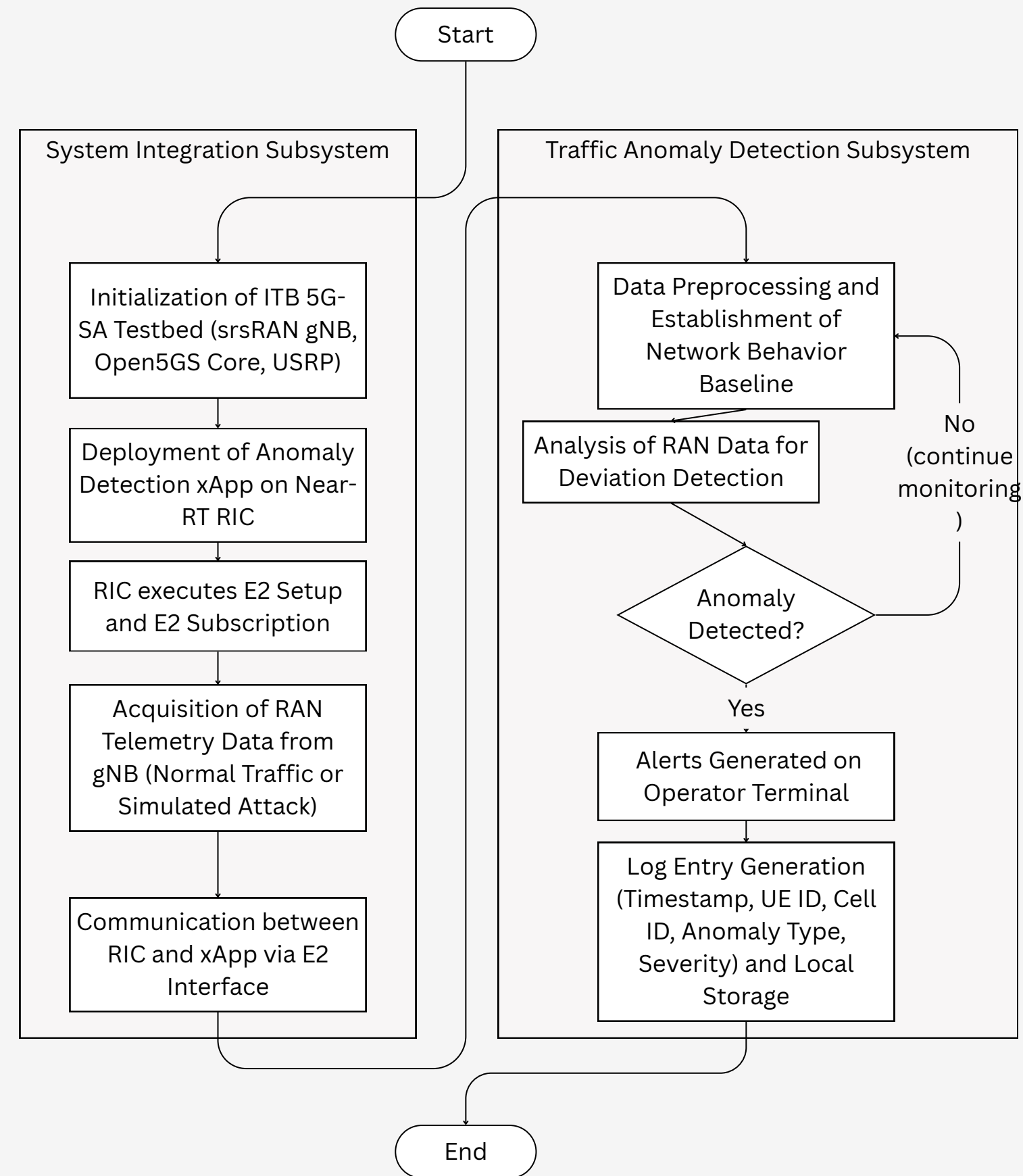
Subobjective Metrics

Subobjective	Measurement Method	Fulfillment Criteria
Subobjective 1: Integration with Near-RT RIC and ITB 5G-SA testbed components.	Deployment verification on the Near-RT RIC platform.	The xApp is successfully deployed and operates continuously without crashing. Telemetry data is transmitted using the E2SM-KPM format via E2AP protocols.
Subobjective 2: High Detection Accuracy.	Evaluation of the ML model against validation datasets.	The model achieves a detection accuracy of 90%. System generates structured logs (Timestamp, UE-ID, Severity) compliant with Cisco IPS and srsRAN standards.
Subobjective 3: High Detection Efficiency.	Resource usage profiling (CPU utilization monitoring).	The xApp-RIC CPU utilization remains < 85% during active operation.
Subobjective 4: Low Latency	Latency measurement of the detection pipeline.	The inference and processing latency for non-critical services does not exceed 100 ms.

Constraint Design Measurement

Subobjective and Constraint	Measurement Method	Fulfillment Criteria
Constraint 1: Does not cause crash, overload, or communication disruption between RIC and gNB.	System stability monitoring during operation.	Zero instances of system overload or disruption to the E2 control plane communication.
Constraint 2: Compatible with srsRAN gNB, Open5GS core, and USRP B205mini-i devices.	Integration testing within the specific hardware/software environment.	Full interoperability with srsRAN, Open5GS, and USRP B205mini-i without library conflicts or connection failures.

Flowchart



Alternative Design Solution

Subsystem	Category	Alternative Design Solution
Subsystem 1: System Integration	RIC Platform	1.O-RAN Software Community RIC 2.FlexRIC
Subsystem 2: Traffic Anomaly Detection	Machine Learning Algorithm	1.Isolation Forest (iForest) 2.Deep Autoencoder 3.LSTM-Autoencoder

No.	Integrated System Alternative Designs
1	OSC RIC - Isolation Forest
2	OSC RIC - Deep Autoencoder
3	OSC RIC - LSTM Autoencoder
4	FlexRIC - Isolation Forest
5	FlexRIC - Deep Autoencoder
6	FlexRIC - LSTM Autoencoder

Platform RIC

Alternative Platform 1: O-RAN

Software Community (OSC) RIC

- Cloud-native reference platform using Kubernetes and microservices.
- (+) Full O-RAN standard compliance (E2AP/E2SM) and high modularity.
- (-) Heavy resource consumption (CPU/RAM) and high deployment complexity

Alternative Platform 2: FlexRIC

- Lightweight, research-oriented platform running as a native process.
- (+) Superior resource efficiency and seamless "out-of-the-box" integration with srsRAN.
- (-) Limited xApp lifecycle management and only partial E2SM standard adherence.

ML Algorithm

Alternative Algorithm 1: Isolation Forest

- Ensemble tree method that explicitly isolates anomalies based on quantitative properties.
- (+) High computational efficiency (linear complexity) and scalability for large datasets.
- (-) Struggles with local anomalies and fails to capture complex inter-feature correlations.

Alternative Algorithm 2: Deep Autoencoder

- Neural network that detects anomalies via reconstruction error in a low-dimensional latent space.
- (+) Superior accuracy for non-linear patterns and effective semi-supervised learning capabilities.
- (-) High computational cost and susceptibility to overfitting on training data.

Alternative Algorithm 3: LSTM-Autoencoder

- Hybrid architecture combining LSTM and Autoencoders to learn temporal dependencies in time-series data.
- (+) Exceptional precision for sequential network data (up to 99.2% accuracy).
- (-) Cons: Requires intensive multi-stage training and lacks transparency ("black box" logic).

Material Cost

Alternative
Solution 1, 2, 3
(OSC-RIC)

List of Component	Description	Total Cost
Mini PC (1 Core)	Standard Mini PC	Rp5,000,000.00
Mini PC (2 Cores + RIC)	High-Performance Mini PC with Large-Capacity RAM (32 GB)	Rp20,000,000.00
USRP B205mini-i + Antenna	SDR Device with Antenna Package	Rp40,000,000.00
Connectivity & Accessories	1× Gigabit Switch, 1× USB 3.0 High-Grade Cable, 1× LAN CAT-6 Cable, Power Management	Rp2,000,000.00
Total Material Cost		Rp51,000,000.00

Alternative
Solution 4, 5, 6
(FlexRIC)

List of Component	Description	Total Cost
Mini PC (1 Core)	Standard Mini PC	Rp5,000,000.00
Mini PC (2 Cores + RIC)	High-Performance Mini PC	Rp10,000,000.00
USRP B205mini-i + Antenna	SDR Device with Antenna Package	Rp40,000,000.00
Connectivity & Accessories	1× Gigabit Switch, 1× USB 3.0 High-Grade Cable, 1× LAN CAT-6 Cable, Power Management	Rp2,000,000.00
Total Material Cost		Rp41,000,000.00

Service Cost

**Alternative
Solution 1, 2, 3
(OSC-RIC)**

List of Component	Description	Total Cost
Application Development Fee	Accumulated application development cost for developers	Rp15.000.000,00
RIC System Installation and Configuration	Installation and Configuration of RIC system	Rp10.000.000,00
Maintenance and Operational Support	Maintenance and operational cost such as electric bill	Rp5.000.000,00
Total Service Cost		Rp30.000.000,00

**Alternative
Solution 4, 5, 6
(FlexRIC)**

List of Component	Description	Total Cost
Application Development Fee	Accumulated application development cost for developers	Rp15.000.000,00
RIC System Installation and Configuration	Installation and Configuration of RIC system	Rp3.000.000,00
Maintenance and Operational Support	Maintenance and operational cost such as electric bill	Rp2.000.000,00
Total Service Cost		Rp20.000.000,00

Financial Analysis

$$NPV = A_0 + \sum_{t=1}^n \frac{F_t}{(1+k)^t}$$

Variables:

A_0 : Costs to be covered

F_t : Selling price per unit

k : Net profit percentage

n : Number of years

Asumptions Taken:

- Three year NPV projection
- Discount rate: 10%
- Profit Margin: Rp24.000.000

Alternative

**Solution 1, 2, 3
(OSC-RIC)**

$$NPV = -Rp81.000.000,00 + \sum_{t=1}^3 \frac{Rp62.000.000,00}{(1+0.1)^t} = Rp73.184.832,00 \quad \textbf{(Positive)}$$

Selling price = Rp65.000.000,00

COGS= Rp41.000.000,00

Alternative

**Solution 4, 5, 6
(FlexRIC)**

$$NPV = -Rp61.000.000,00 + \sum_{t=1}^3 \frac{Rp67.000.000,00}{(1+0.1)^t} = Rp105.619.081,00 \quad \textbf{(Positive)}$$

Selling price = Rp75.000.000,00

COGS= Rp51.000.000,00

Solution Design Determination

Weighted Category Matrix

Category	Weight
Near-RT RIC Platform	3
ML Algorithm	2
Financial Analysis	1

Decision Matrix

Categories Alternative	Near-RT RIC Platform (Weight: 3)	ML Algorithm (Weight: 2)	Financial Analysis (Weight: 1)	Total Score
Alternative 1	2	3	3	15
Alternative 2	2	4	3	17
Alternative 3	2	5	3	19
Alternative 4	5	3	5	26
Alternative 5	5	3	5	28
Alternative 6	5	5	5	30

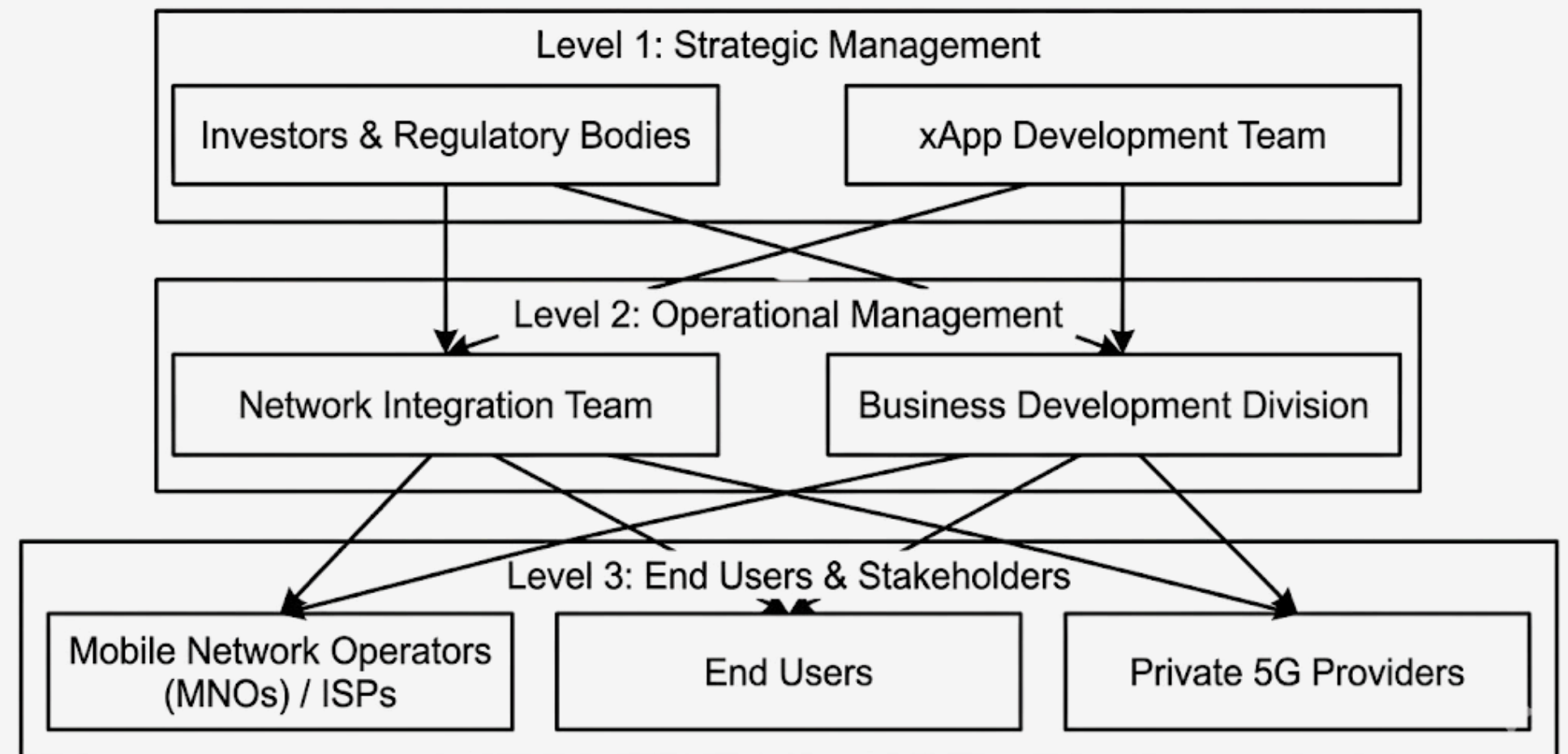
Chosen Alternative: Alternative 6 (FlexRIC - LSTM Autoencoder)

Project Design Planning

Funding Sources and Acquisition Strategies

- Initial funding will be provided by Telecommunication Engineering ITB with laboratory facilities, such as mini PC, USRP, and UE.
- Advanced development and prototype implementation requires a higher scale of funding from external investors.

Organizational Structure



Plans for Final Project II

No	Activity	Description
1	Installation of FlexRIC and Testbed Setup	Conducting the installation of the FlexRIC platform within the designated testbed environment.
2	Development of Cyberattack Scenarios and Dataset Collection	Designing cyberattack simulation scenarios and performing data collection.
3	Data Preprocessing and Initial Model Training	Executing initial preprocessing of raw data and developing the first anomaly detection model using established algorithms.
4	Deep Learning Model Optimization	Enhancing and fine-tuning the anomaly detection model using deep learning methods (LSTM-Autoencoder).
5	Implementation of xApp Code with FlexRIC SDK	Developing xApp programming using the Python language and the FlexRIC SDK, with emphasis on algorithm integration.
6	Integration of ML Models into FlexRIC	Integrating the completed Machine Learning model into the FlexRIC system.
7	Comprehensive Testing and Performance Measurement	Conducting system-level testing to evaluate and measure performance parameters according to established evaluation metrics.
8	Final Report Preparation and Revision	Preparing the final thesis report and conducting necessary revisions.

**Thank You
for Your
Attention**

